Trading Strategy Robustness Report

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Overview of Robustness Metrics

Robustness is crucial to evaluate when backtesting trading strategies. This is necessary to ensure that they perform consistently over time and are able to withstand various market conditions. Just analyzing a strategy's average returns and profitability is insufficient, as it does not properly account for potential risks and drawdowns that you would care about if you were to adopt the trading strategy in the long-term. By measuring the robustness metrics such as Sharpe Ratio and Sortino Ratio, we gain a better understanding of the strategy's risk-adjusted performance, its ability to recover from losses, and its resilience during higher market volatility.

To measure the robustness of our trading strategies, we will measure the Sharpe Ratio and the Sortino Ratio of the returns for each currency for different periods of time. The Sharpe Ratio measures the risk-adjusted returns by dividing the average return by the standard deviation of the returns.

$$Sharpe\ Ratio = \frac{R_p - R_f}{\sigma_p}$$

where:

R_p = return of portfolio

R_f = risk-free rate

σ_p = standard deviation of the portfolio's excess return

Figure 1: Sharpe Ratio Formula

The Sortino Ratio is similar to the Sharpe Ratio, but only considers the downside deviation (negative returns) in the risk calculation. This helps us evaluate the robustness in avoiding large losses.

$$\text{Sortino Ratio} = \frac{R_p - r_f}{\sigma_d}$$

where:

R_p = Actual or expected portfolio return

r_f = Risk-free rate

σ_d = Standard deviation of the downside

Figure 2: Sortino Ratio Formula

We will now calculate the Sharpe Ratio and Sortino Ratio to measure the robustness of the RSI trading strategy.

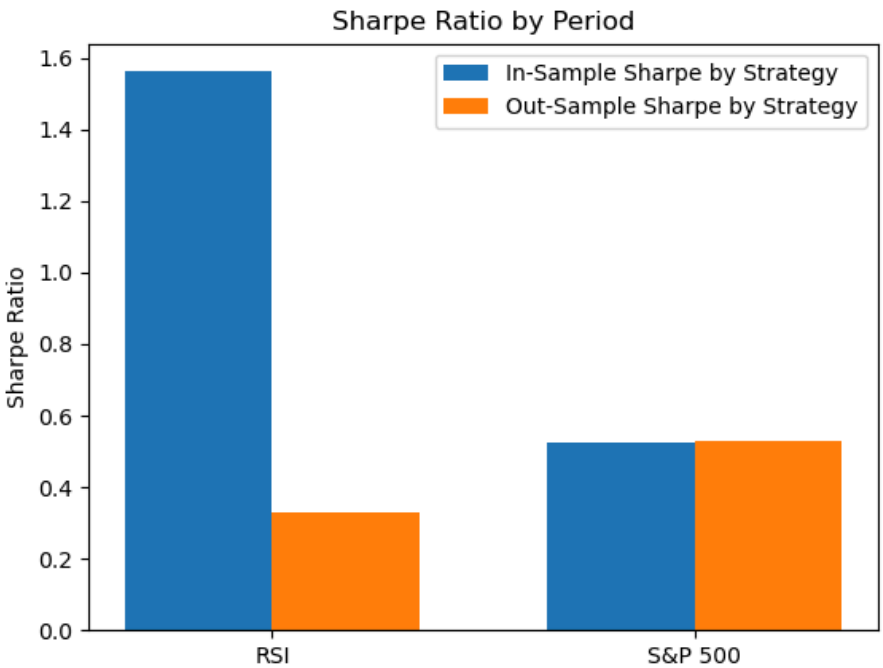
Strategy Performance

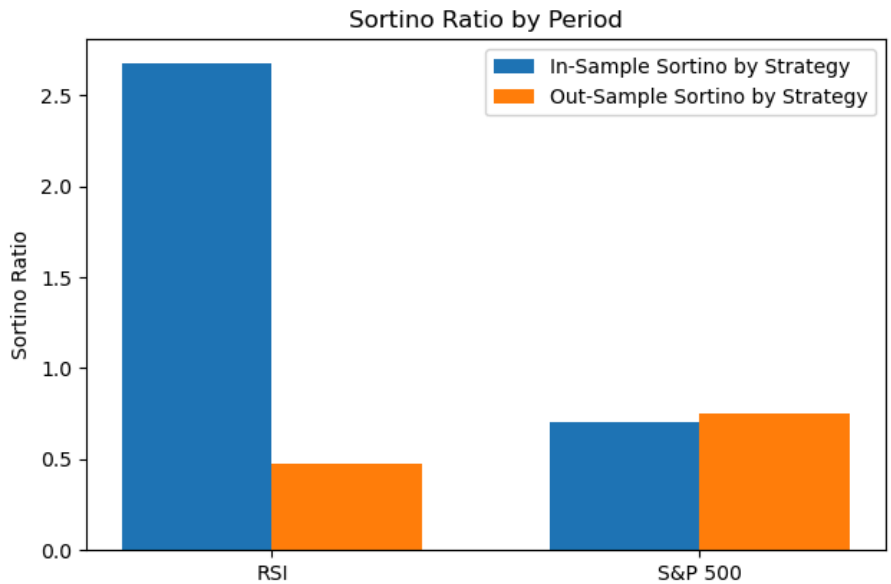
In the table below, we show the Sharpe Ratio and the Sortino Ratio for the RSI trading strategy, the S&P 500, and the RSI trading strategy for individual currencies for the time period from 05-01-2019 to 02-02-2021 (labeled as in-sample), and from 02-03-2021 to 02-27-2024

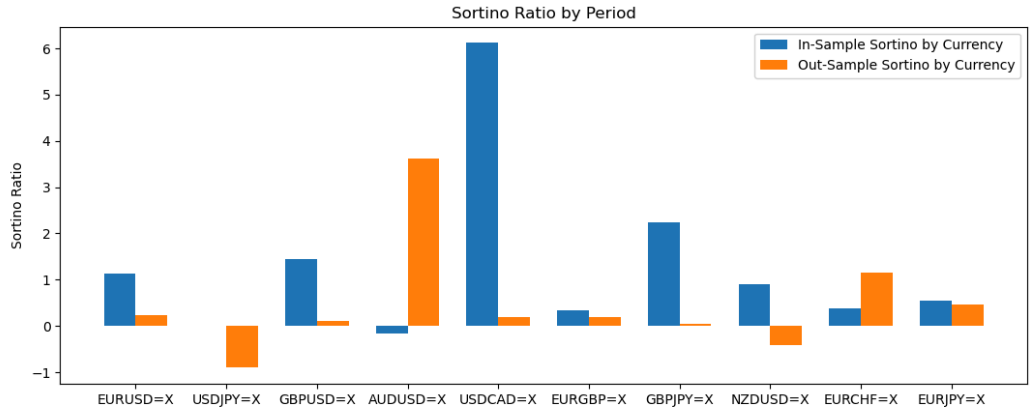
	Sharpe Ratio In-Sample	**Sharpe Ratio Out-of-Sample**	**Sortino Ratio In-Sample**	**Sortino Ratio Out-of-Sample**
S&P 500	0.53	0.53	0.70	0.47
Total RSI	1.56	0.33	2.67	0.75
EURUSD RSI	0.63	0.16	1.14	0.23
USDJPY RSI	Inf (no trades)	-0.68	Inf (no trades)	-0.90
GBPUSD RSI	0.76	0.08	1.45	0.12
AUDUSD RSI	-0.11	1.35	-0.17	3.62
USDCAD RSI	1.57	0.13	6.12	0.19

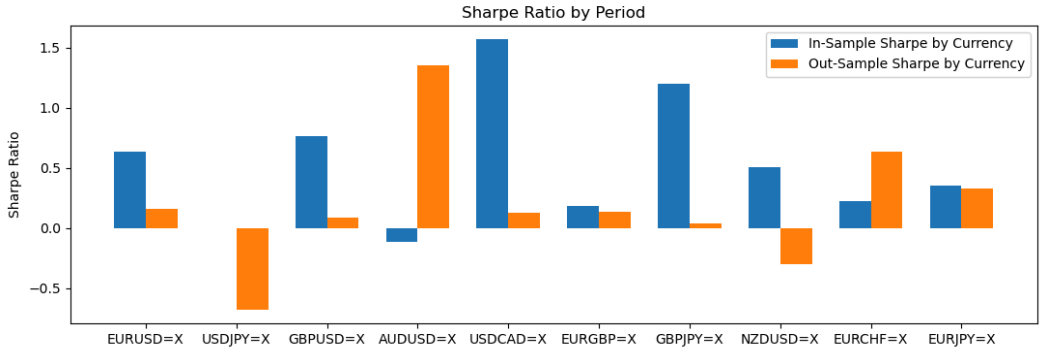
EURGBP RSI	0.18	0.14	0.33	0.18
GBPJPY RSI	1.20	0.04	2.25	0.05
NZDUSD RSI	0.51	-0.30	0.91	-0.41
EURCHF RSI	0.23	0.64	0.39	1.15
EURJPY RSI	0.35	0.33	0.54	0.46

A higher Sharpe Ratio and a higher Sortino Ratio means that the strategy has better risk-adjusted returns. So, we see that the RSI trading strategy has a much better in-sample Sharpe and Sortino ratio than the S&P 500, though out-of-sample has a worse Sharpe Ratio and a better Sortino Ratio. We also see that certain currencies have better risk-adjusted returns than others. For example, EUR/CHF gives good returns both in-sample and out-of-sample, though NZD/USD is good in-sample but actually gives us negative returns out-of-sample.


Sharpe Ratio by Period
In-Sample Sharpe by Strategy
Out-Sample Sharpe by Strategy
Sharpe Ratio
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
RSI
S&P 500


Sortino Ratio by Period
In-Sample Sortino by Strategy
Out-Sample Sortino by Strategy
Sortino Ratio
2.5
2.0
1.5
1.0
0.5
0.0
RSI
S&P 500





Conclusion

Our results show that our RSI-based Trading Strategy does have good risk-adjusted returns. Specifically, it does even better than the S&P 500 at getting returns and avoiding large losses, as shown by the better Sortino Ratio. However, the strategy's Sharpe and Sortino Ratio are not consistently positive for every currency tested on or for every time period tested on, so it could likely be more robust. If using this strategy, traders should be careful to avoid large losses because this test of robustness shows that it is possible this trading strategy is not going to work in every situation and through every market conditions, though it may not be extremely volatile overall.

References

Fernando, Jason. "Sharpe Ratio: Definition, Formula, and Examples." *Investopedia*, Investopedia, www.investopedia.com/terms/s/sharperatio.asp. Accessed 27 Mar. 2024.

Kenton, Will. "Sortino Ratio: Definition, Formula, Calculation, and Example." *Investopedia*, Investopedia, www.investopedia.com/terms/s/sortinoratio.asp. Accessed 27 Mar. 2024.

Code

https://github.com/jacksonmeehan/AQFAAT-RSI-backtest/tree/5fa56be1ac6b80e1eb1d2dc5dd65f7585db43d66